Lecture 21: Section 5.6

The Fundamental Theorem of Calculus

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The Big Idea

The Fundamental Theorem of Calculus (FTC) is one of the most important ideas in mathematics. It connects two seemingly different concepts:

- **Differentiation** (finding slopes/tangent lines)
- **Integration** (finding areas under curves)

Before FTC: We calculated derivatives and integrals separately, using different rules.

After FTC: We realize these are *inverse operations* - like addition/subtraction or multiplication/division!

FTC Part 1: The Statement

Theorem (Fundamental Theorem of Calculus, Part 1)

If f is continuous on $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a).$$

In Simple Terms

To calculate a definite integral:

1. Find an antiderivative $F(x)$ of $f(x)$
2. Evaluate F at the endpoints: $F(b) - F(a)$

This is HUGE! Before FTC, calculating areas was hard (limits of Riemann sums). After FTC Part 1, we have an algebraic method!

Worked Example: FTC Part 1

Example

Calculate $\int_1^4 (3x^2 - 2x + 1)dx$.

Solution:

1. Find an antiderivative: $F(x) = x^3 - x^2 + x$ (since $F'(x) = 3x^2 - 2x + 1$)
2. Apply FTC Part 1:

$$\begin{aligned}\int_1^4 (3x^2 - 2x + 1)dx &= F(4) - F(1) \\ &= \left[4^3 - 4^2 + 4\right] - \left[1^3 - 1^2 + 1\right] \\ &= (64 - 16 + 4) - (1 - 1 + 1) \\ &= 52 - 1 = 51\end{aligned}$$

Important Caveat: Function Must Be Continuous!

Example (When FTC Part 1 Fails)

Consider $\int_{-1}^{1} \frac{1}{x^2} dx$.
Attempt using FTC:

- Antiderivative: $F(x) = -\frac{1}{x}$ (since $F'(x) = \frac{1}{x^2}$)
- Apply FTC: $F(1) - F(-1) = (-1) - (1) = -2$

But this is incorrect! $f(x) = \frac{1}{x^2}$ is NOT continuous on $[-1, 1]$ (it has a vertical asymptote at $x = 0$).
The area under this curve from -1 to 1 is actually **infinite** (the integral diverges).

Key Point: FTC Part 1 requires f to be continuous on the entire interval $[a, b]$.

Total Area vs. Net Area

Example

Consider $\int_0^2 (1 - x^2)dx$.

Net Area (Using FTC Part 1):

$$\int_0^2 (1 - x^2)dx = \left[x - \frac{x^3}{3}\right]_0^2$$
$$= \left(2 - \frac{8}{3}\right) - (0) = -\frac{2}{3}$$

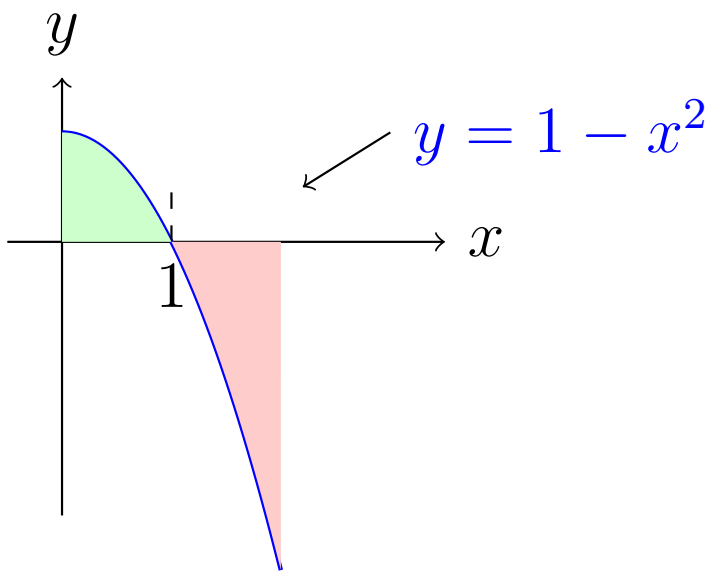
Net area is negative because more area is below the x-axis.

Total Area:

- Find where $1 - x^2 = 0$: $x = 1$ (in $[0, 2]$)
- Split at $x = 1$:

$$\text{Total Area} = \left| \int_0^1 (1 - x^2) dx \right| + \left| \int_1^2 (1 - x^2) dx \right|$$
$$= \left| \frac{2}{3} \right| + \left| -\frac{4}{3} \right| = \frac{2}{3} + \frac{4}{3} = 2$$

Visualizing Total vs. Net Area



FTC Part 2: The Statement

Theorem (Fundamental Theorem of Calculus, Part 2)

Let f be continuous on $[a, b]$, and define F by

$$F(x) = \int_a^x f(t)\,dt \quad \text{for } a \leq x \leq b.$$

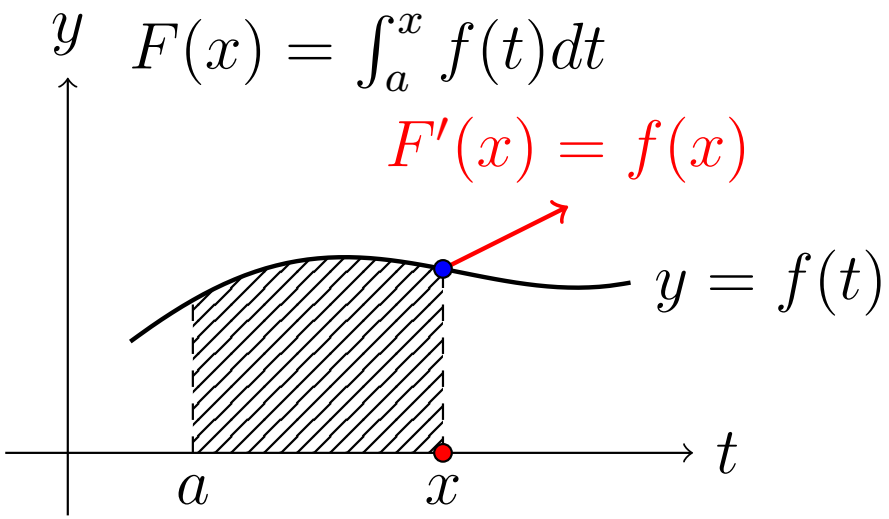
Then F is differentiable on (a, b), and

$$F'(x) = \frac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x).$$

Extended version: If $F(u) = \int_a^u f(t)\,dt$, then $\frac{d}{dx}F(u(x)) = f(u(x)) \cdot u'(x)$.

Visualizing FTC Part 2

- Start with a continuous function f
- Create a new function $F(x) =$ "area under f from a to x"
- The derivative of this area function is the original function f!



Key insight: As x moves to the right, the area $F(x)$ accumulates at a rate equal to the height $f(x)$ of the function.

Worked Example: FTC Part 2

Example

Find $G'(x)$ for:

$$G(x) = \int_2^{x^2} \frac{t}{1+t^3}\, dt$$

Solution: By the Fundamental Theorem of Calculus Part 2 and the chain rule: If $F(u) = \int_a^u f(t)\, dt$, then $\frac{d}{dx}F(u(x)) = f(u(x)) \cdot u'(x)$. Here, $f(t) = \frac{t}{1+t^3}$ and $u(x) = x^2$.

$$G'(x) = \frac{x^2}{1+(x^2)^3} \cdot \frac{d}{dx}(x^2) = \frac{x^2}{1+x^6} \cdot 2x = \frac{2x^3}{1+x^6}$$

Indefinite vs. Definite Integrals

Definition (Indefinite Integral)

The **indefinite integral** of f is the family of all antiderivatives:

$$\int f(x)\,dx = F(x) + C$$

where $F'(x) = f(x)$ and C is an arbitrary constant.

Definition (Definite Integral)

The **definite integral** of f from a to b is a number:

$$\int_a^b f(x)\,dx = \text{net signed area under } f \text{ from } a \text{ to } b$$

Connecting the Two

From FTC Part 2: If $F(x) = \int_a^x f(t)dt$, then $F'(x) = f(x)$.
So F is an **antiderivative** of f!
But wait... the indefinite integral $\int f(x)dx$ represents **all** antiderivatives.
This suggests:

$$\int_a^x f(t)dt \text{ is one specific antiderivative of } f$$

Which one? The one that equals 0 when $x = a$ (since $\int_a^a f(t)dt = 0$).
So:

- **Indefinite integral:** All antiderivatives: $F(x) + C$
- **Definite integral (as a function of upper limit):** One particular antiderivative: $\int_a^x f(t)dt$

What's a Dummy Variable?

Consider: $\int_a^x f(t)dt$ vs. $\int_a^x f(u)du$ vs. $\int_a^x f(\xi)d\xi$

Definition

A **dummy variable** (or bound variable) is a variable of integration that appears in a definite integral. It "disappears" after integration, so its name doesn't matter.

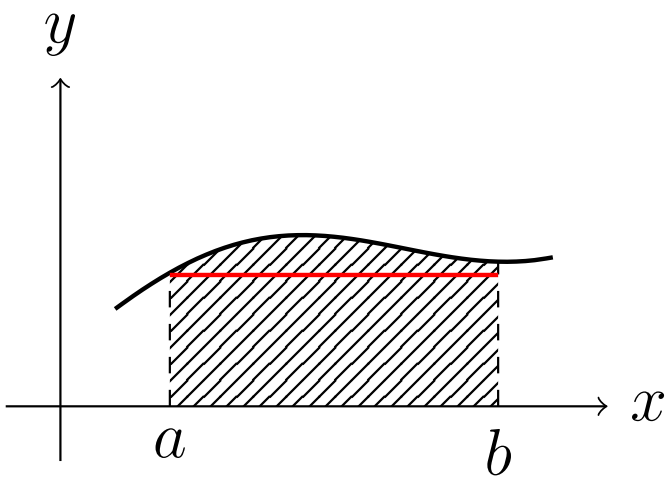
Example

All these are the same:

- $\int_0^2 x^2 dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}$
- $\int_0^2 t^2 dt = \left[\frac{t^3}{3}\right]_0^2 = \frac{8}{3}$
- $\int_0^2 u^2 du = \left[\frac{u^3}{3}\right]_0^2 = \frac{8}{3}$

Average Value of a Function

What's the "average height" of a function over an interval?



$$\text{Area} = \int_a^b f(x)dx = f_{\text{avg}} \times (b-a)$$

So: $f_{\text{avg}} = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)dx$

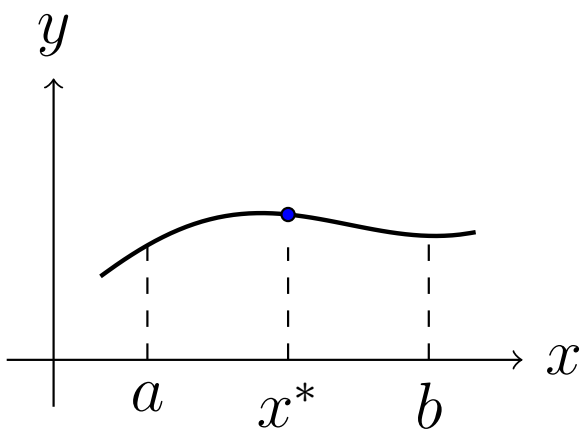
Mean Value Theorem for Integrals

Theorem (Mean Value Theorem for Integrals)

If f is continuous on $[a, b]$, then there exists at least one number x^* in (a, b) such that

$$\int_a^b f(x)dx = f(x^*)(b - a)$$

That is, f attains its average value at some point x^* in the interval.



Example: MVT for Integrals

Example

Find x^* in $[1, 4]$ such that $f(x^*) = f_{\text{avg}}$ for $f(x) = x^2$.

Solution:

1. Calculate the average value:

$$f_{\text{avg}} = \frac{1}{4-1}\int_1^4 x^2 dx = \frac{1}{3}\left[\frac{x^3}{3}\right]_1^4 = \frac{1}{3}\left(\frac{64}{3} - \frac{1}{3}\right) = \frac{63}{9} = 7$$

2. Find x^* such that $f(x^*) = (x^*)^2 = 7$:

$$x^* = \sqrt{7} \approx 2.645$$

(Note: $\sqrt{7}$ is indeed in $(1, 4)$)

So at $x^* = \sqrt{7}$, the function value equals the average value over $[1, 4]$.

Inverse Operations

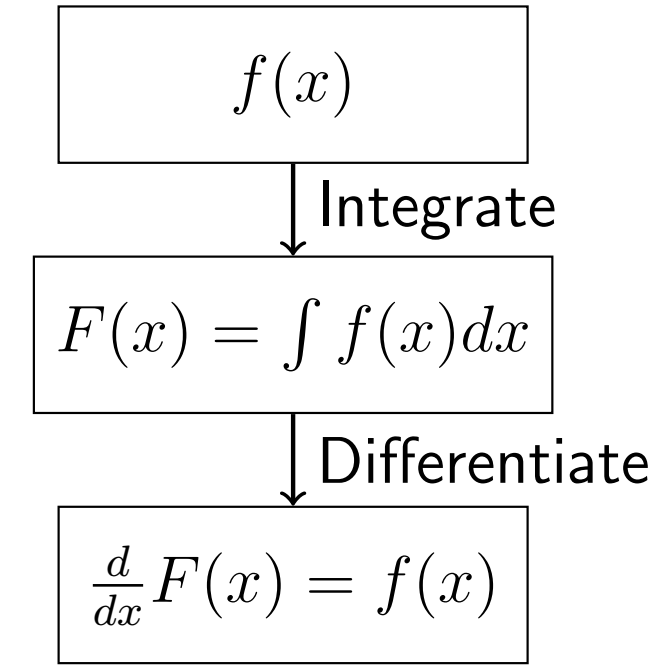
Think about familiar inverse operations:

- Addition $\leftrightarrow$ Subtraction: $(x + 3) - 3 = x$
- Multiplication $\leftrightarrow$ Division: $(x \times 3) \div 3 = x$
- Exponentiation $\leftrightarrow$ Logarithms: $\log_b(b^x) = x$

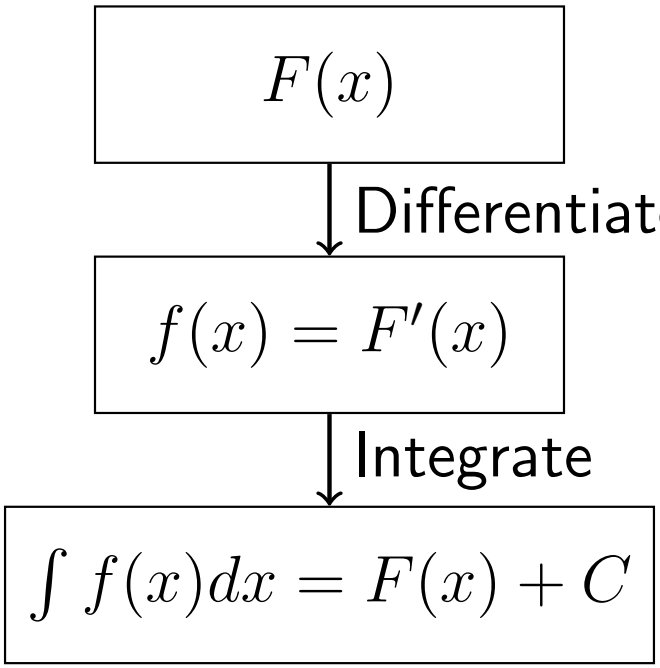
Now we have a new pair: **Differentiation** $\leftrightarrow$ **Integration**
From FTC:

- If you integrate then differentiate: $\frac{d}{dx}\left[\int_a^x f(t)dt\right] = f(x)$ (FTC Part 2)
- If you differentiate then integrate: $\int_a^x F'(t)dt = F(x) - F(a)$ (FTC Part 1)

Visualizing Inverse Operations



We get back to $f(x)$!



We get back to $F(x)$ (plus constant)!

A Powerful Application

Definition (Integrating Rates of Change)

If $R(t)$ represents a rate of change (velocity, growth rate, flow rate, etc.), then

$$\int_a^b F'(x)dt = F(b) - F(a)$$

This follows directly from FTC Part 1.

Examples:

- **Velocity to Displacement:** $\int_a^b v(t)dt = s(b) - s(a)$
- **Growth Rate to Total Growth:** $\int_a^b P'(t)dt = P(b) - P(a)$
- **Marginal Cost to Total Cost:** $\int_a^b C'(x)dx = C(b) - C(a)$
- **Flow Rate to Total Volume:** $\int_a^b r(t)dt = V(b) - V(a)$

Worked Examples

Example

Water flows into a tank at a rate of $r(t) = 2t + 5$ liters/minute. How much water flows in during the first 3 minutes?

Solution: Total water $= \int_0^3 r(t)dt = \int_0^3 (2t + 5)dt$

$$= \left[t^2 + 5t\right]_0^3 = (9 + 15) - (0 + 0) = 24 \text{ liters.}$$

Worked Examples

Example

The marginal cost to produce x items is $C'(x) = 0.3x^2 - 2x + 5$. Find the total cost increase from producing 10 to 20 items.

Solution: Total cost increase $=$
$\int_{10}^{20} C'(x)dx = \int_{10}^{20}(0.3x^2 - 2x + 5)dx$

$$= \left[0.1x^3 - x^2 + 5x\right]_{10}^{20} = (800 - 400 + 100) - (100 - 100 + 50) = 450$$

Practice Problems: FTC Part 1

1. Evaluate $\int_0^2 (3x^2 - 4x + 1)dx$ using FTC Part 1.
2. Evaluate $\int_{-1}^1 (x^3 - 2x)dx$.
3. Find $\int_0^{\pi/2} \cos x dx$.
4. Evaluate $\int_1^4 \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) dx$.
5. A car's velocity is $v(t) = 60 - 3t$ mph. How far does it travel from $t = 0$ to $t = 10$ hours?

Practice Problems: FTC Part 2

1. Find $F'(x)$ if $F(x) = \int_0^x \sin(t^2)dt$.
2. Find $G'(x)$ if $G(x) = \int_1^{x^2} \sqrt{1+t^3}dt$. (Hint: Chain rule!)
3. If $H(x) = \int_x^3 e^{-t^2}dt$, find $H'(x)$. (Hint: Reverse limits first)
4. Use FTC Part 2 to find the derivative of $\int_\pi^x \cos(t)dt$.

More Practice Problems

1. Find the average value of $f(x) = x^2$ on $[0, 2]$.
2. Find x^* in $[0, 2]$ such that $f(x^*) = f_{\text{avg}}$ for $f(x) = x^2$.
3. True or False: $\int_a^b f(x)dx = \int_a^b f(t)dt$.
4. If $f(x) = \frac{d}{dx}\int_0^x g(t)dt$, what is $f'(x)$?
5. Find both the net area and total area for $\int_0^3 (x-2)dx$.

Key Takeaways

- **FTC Part 1:** To evaluate $\int_a^b f(x)dx$, find any antiderivative F and compute $F(b) - F(a)$.
- **FTC Part 2:** Differentiating an integral (with variable upper limit) gives back the original function.
- **Dummy variables** disappear after integration - only the limits matter in the final answer.
- The **Mean Value Theorem for Integrals** guarantees $f(x^*) = f_{\text{avg}}$ for some x^* in (a, b).
- **Differentiation and integration** are inverse processes.
- **Integrating a rate of change** gives the total change.
- **Net area** considers sign, while **total area** sums absolute values.